

11. The combinational circuit represents



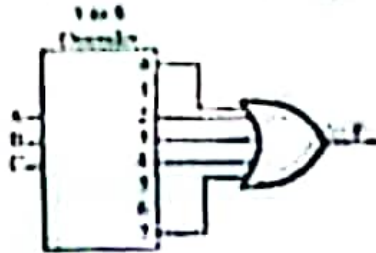
- (a) Decoder
- (b) Encoder
- (c) **Comparator**
- (d) Multiplexer

12. The function represented by the following 4X1 Multiplexer is



- (a) $F = x'yz + x'y'z + xy$
- (b) $F = xy + xz + x'y'z$
- (c) $F = x'y'z + x'yz + xy$
- (d) $F = x'y'z + xy'z + x'y$

13. Using the following Decoder, the simplified SOP for the function F is:



- (a) **$F = B'C' + BC + AB'$**
- (b) $F = B'C + BC + AB'$
- (c) $F = BC' + BC + AB'$
- (d) $F = B'C' + BC + A'B$

14. To implement $A \oplus B$ using NOR with three inputs, the third input should be

- a) Ground
- b) To one of inputs (A,B)
- c) +5V
- d) Answers b or c

15. Which one of these called universal gate:

- a) NAND only
- b) **NAND, NOR**
- c) AND, OR, NOT
- d) OR, AND, NOT, NOR, XOR, XNOR, NAND

16. What is the output of NOR with two inputs A,B, if A is connected to +5v;

- a) Always zero
- b) Always one
- c) 0
- d) **1**

Question 1: Multiple choice Question (30 marks)

1. A combinational circuit that selects one from many inputs are _____
(a) Encoder (b) Decoder (c) Demultiplexer (d) Multiplexer
2. A combinational circuit is one in which the output depends on the _____
(a) Input combination at the time
3. (b) Input combination and the previous output
(c) Input combination at that time and the previous input combination
(d) Present output and the previous output
4. In a multiplexer the output depends on its _____
(a) Data inputs (b) Select inputs (c) Select outputs (d) Enable pin
5. In 1-to-4 multiplexer, if $C1 = 0$ & $C2 = 1$, then the output will be _____
(a) Y_0 (b) Y_1 (c) Y_2 (d) Y_3
6. In 1-to-4 multiplexer, if $C1 = 1$ & $C2 = 1$, then the output will be _____
(a) Y_0 (b) Y_1 (c) Y_2 (d) Y_3
7. Which of the following is not a sequential circuit?
(a) Flip-flop (b) Counter (c) Shift Register (d) Multiplexer
8. The sequential circuit the output state depend upon _____
(a) Past input state (b) Present input state
(c) Present as well as past input (d) None of the above
9. A 4 bit counter will count _____
(a) 0-3 (b) 0-4 (c) 0-15 (d) 0-16
10. The following truth table represents a _____ gate

X	Y	F
0	0	1
0	1	0
1	0	0
1	1	1

(a) NOR
(b) HALF ADDER
(c) XNOR
(d) XOR